Unit Reflective IR

SKU:U175

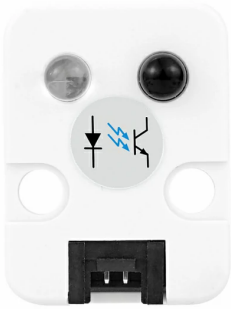












Description

Unit Reflective IR is a self-emitting and receiving infrared sensor unit, mainly used to detect the reflected infrared light of a target object, and outputs detection status via both analog and digital signals. Its unique design incorporates an onboard adjustable potentiometer for tuning the sensitivity of the digital output, allowing users to adjust the sensor's parameters as needed for increased flexibility and precision. This product is suitable for applications such as object detection, line tracking, distance measurement, and automation systems.

Note: The rear potentiometer adjusts detection distance (sensitivity). It is pre-set at the factory to a comparator threshold voltage of 1.65V. Do not adjust the potentiometer unless necessary. Valid voltage range: 0.3V ≥ 1.65V ≤ 2.15V. Beyond this range — when > 2.15V, the red LED will stay on; when < 0.3V, the red LED will stay off, causing detection failure. (The potentiometer voltage can be measured at the comparator LM393 inverting terminal according to the schematic.)

Features

- **Infrared sensing technology**: Utilizes infrared sensing to efficiently and accurately detect reflected IR light from target objects.
- **Analog signal output**: The IR receiver and 10k resistor divider generate a voltage corresponding to the intensity of the reflected IR signal from the obstacle.

- **Digital signal output**: Integrated LM393 comparator converts the analog signal into a reliable digital signal for easy processing in digital systems.
- **LED indicator**: Digital output controls the LED indicator, visually indicating detection status for quick user feedback.
- **Adjustable potentiometer**: Equipped with a 10k adjustable potentiometer to fine-tune sensitivity and threshold, optimizing performance for different environments and applications.
- **Development platforms**: Arduino, UIFlow, etc.

Includes

- 1 x Unit Reflective IR
- 1 x HY2.0-4P Grove Cable (20cm)
- 1 x Potentiometer screwdriver

Applications

- Object detection
- Distance measurement
- Automation systems
- Security systems
- Line tracking

Specifications

Specification	Parameter
Analog Output Range	0-4096 (ESP32@12bit)
Digital Output	TTL
Operating Voltage	5V
Operating Temp.	0 ~ 40°C
Product Size	32.0 x 24.0 x 12.0mm
Product Weight	4.6g
Package Size	138.0 x 93.0 x 13.0mm
Gross Weight	14.6g

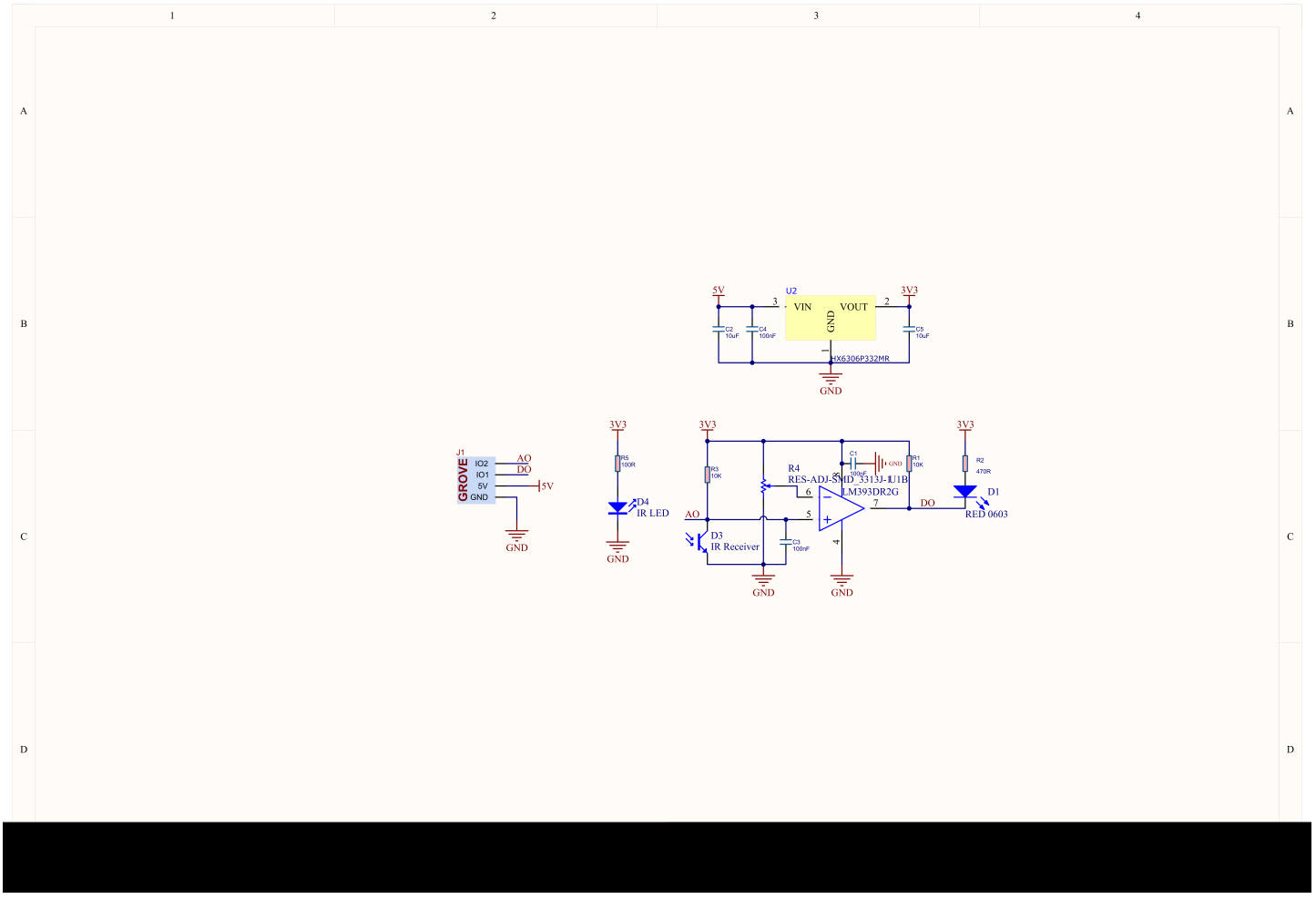
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Detection Range

	Threshold Voltage: 1.65V (factory default, potentiometer centered)	Threshold Voltage: 2.15V
Black object max range	0cm (Test object: 100% black printed A4 paper)	6.5cm (Test object: 100% black printed A4 paper)
White object max range	13cm (Test object: white A4 paper)	28cm (Test object: white A4 paper)
Color object max range	14cm (Test object: magazine cover, blue/green/yellow)	25.5cm (Test object: magazine cover, blue/green/yellow)
Glossy object max range	31cm (Test object: glossy metal mesh)	43cm (Test object: glossy metal mesh)

Schematics

- Unit Reflective IR Schematics PDF

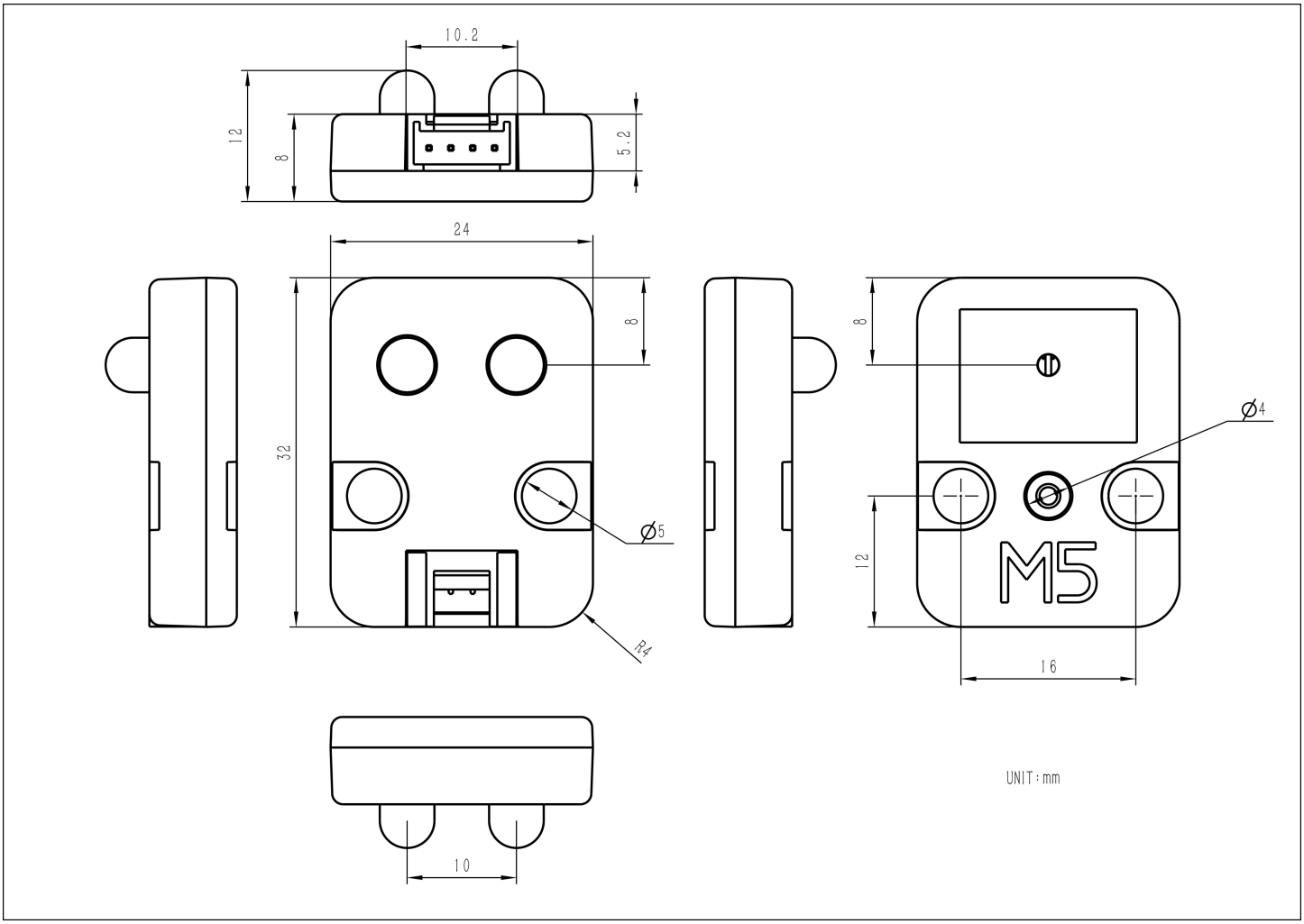


PinMap

Unit Reflective IR

HY2.0-4P	Black	Red	Yellow	White
PORT.B	GND	5V	Digital Output	Analog Output

Model Size



Structure

- Unit Reflective IR Structure Files

Softwares

Arduino

- Unit Reflective IR Arduino Tutorial

UiFlow1

- Unit Reflective IR UiFlow1 Docs

UiFlow2

- Unit Reflective IR UiFlow2 Docs

Video

- Unit Reflective IR Example

U175 Reflective IR Unit 视频.mp4

- Unit Reflective IR UiFlow Usage

Product Comparison

UNIT	SKU	NOTE
IR Unit	U002	Supports encoding/decoding
Reflective IR Unit	U175	Self-emitting/receiving, no encode/decode